DAMIEN CHEMISTRY OLYMPIAD NOTES

Atomic Structure

Question Bank with Worked Answers

Core set: 24 concepts × 3 levels = 72 questions
Extension set: 6 mass-spectrometry and 8 photoelectron-spectroscopy questions

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1. Atomic Composition

1.1 Proton, neutron and electron numbers in atoms and ions

1A Easy

An ion has the symbol $^{27}_{13}Al^{3+}$.

Determine the number of:

1. protons,
2. neutrons,
3. electrons.

Answer

$N(\text{protons}) = 13$, $N(\text{neutrons}) = 27 - 13 = 14$,
$N(\text{electrons}) = 13 - 3 = 10$.

1B Medium

An ion X^{2-} contains 18 electrons and 16 neutrons.

Determine:

1. the proton number of X,
2. the nucleon number of the ion,
3. the identity of X.

Answer

Because the ion has a 2− charge,

$$N(\text{electrons}) = N(\text{protons}) + 2.$$

Hence,

$$N(\text{protons}) = 18 - 2 = 16.$$

The element with proton number 16 is sulfur. Its nucleon number is

$$A = 16 + 16 = 32.$$

Therefore, the ion is

$$^{32}_{16}S^{2-}$$

1C Hard

The ion X^{2+} contains 18 electrons and 22 neutrons. The ion Y^- has the same nucleon number as X^{2+} and contains 20 electrons. Determine the nuclide notation of both ions.

Answer

For X^{2+} ,

$$Z_X = 18 + 2 = 20.$$

Thus X is calcium, and

$$A_Z = 20 + 22 = 42.$$

Therefore, $^{42}_{20}Ca^{2+}$.

For Y^-

,

$$Z_Y = 20 - 1 = 19.$$

Thus Y is potassium. Since its nucleon number is also 42, $^{42}_{19}K^{-1}$

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1.2 Proton number, nucleon number and isotopes

2A Easy

Define:
1. proton number,
2. nucleon number,
3. isotope.

Answer

1. Proton number: the number of protons in the nucleus of an atom.
2. Nucleon number: the total number of protons and neutrons in the nucleus.
3. Isotopes: atoms of the same element with the same number of protons but different numbers of neutrons.

2B Medium

Two atoms are represented by ${}^{63}_{29}\mathrm{X}$ and ${}^{65}_{29}\mathrm{X}$.
Explain:
1. why they are isotopes,
2. which subatomic particle accounts for their different masses,
3. why their chemical properties are very similar.

Answer

Both atoms contain 29 protons, so they are atoms of the same element. Their nucleon numbers differ, so they are isotopes. Their neutron numbers are

$$63 - 29 = 34 \text{ and } 65 - 29 = 36.$$

Thus, neutrons account for the difference in mass. Neutral atoms of both isotopes contain 29 electrons and therefore have the same electronic configuration. Since chemical properties depend primarily on electronic configuration, their chemical properties are very similar.

2C Hard

Two neutral species P and Q satisfy the following conditions:
- P and Q have the same number of neutrons.
- P has one more proton than Q.
- P has a nucleon number one greater than Q.

Are P and Q isotopes? Explain whether they are likely to have similar chemical properties.

Answer

They are not isotopes because isotopes must have the same number of protons. Let Q have Z protons and N neutrons. Then

$$A_Q = Z + N.$$

Species P has $Z + 1$ protons and the same N neutrons, so

$$A_P = (Z + 1) + N = A_Q + 1.$$

The information is therefore internally consistent, but P and Q are different elements. Their neutral atoms have different numbers of electrons and different electronic configurations, so they are not necessarily chemically similar.

1.3 Relative charge and relative mass

3A Easy

State the relative charge and relative mass of a proton, neutron and electron.

Answer

Particle	Relative charge	Relative mass
Proton	+1	1
Neutron	0	1
Electron	−1	approximately $\frac{1}{1840}$

3B Medium

Arrange the following particles in increasing magnitude of charge-to-mass ratio: ${}^{4}_{2}\mathrm{He}^{2+}$, ${}^{1}_{1}\mathrm{H}^{+}$, ${}^{2}_{1}\mathrm{H}^{+}$, e^{-}. Explain your ordering.

Answer

Using relative values,

$$\left|\left(\frac{q}{m}\right)_{{}^{4}\mathrm{He}^{2+}}\right| = \frac{2}{4} = \frac{1}{2}, \quad \left|\left(\frac{q}{m}\right)_{{}^{2}\mathrm{H}^{+}}\right| = \frac{1}{2},$$

whereas

$$\left|\left(\frac{q}{m}\right)_{{}^{1}\mathrm{H}^{+}}\right| = 1.$$

The electron has a charge magnitude of 1 but a very small mass, so it has by far the greatest charge-to-mass ratio. Therefore,

$$\boxed{{}^{4}\mathrm{He}^{2+} = {}^{2}\mathrm{H}^{+} < {}^{1}\mathrm{H}^{+} < e^{-}}$$

3C Hard — **Multi-statement MCQ**

A proton, an electron and an alpha particle are placed separately in the same uniform electric field.
1. The alpha particle has a smaller charge-to-mass ratio than the proton.
2. The proton and electron experience accelerations of equal magnitude.
3. The particle with the largest charge-to-mass ratio experiences the largest acceleration magnitude.

Which statements are correct? A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

Answer

In a uniform electric field,

$$a = \frac{F}{m} = \frac{qE}{m}$$

Thus, acceleration magnitude depends on the magnitude of the charge-to-mass ratio. Statement 1 is correct because the alpha particle has $|q|/m = 2/4 = 1/2$, whereas the proton has $|q|/m = 1$. Statement 2 is incorrect because the electron's much smaller mass gives it a much greater acceleration magnitude. Statement 3 is correct. Therefore, the answer is D.

1.4 Behaviour of charged particles in electric fields

4A Easy

A positively charged ion enters a uniform electric field directed from left to right.
State:
1. the direction of the force on the ion,
2. the direction of the force on an electron in the same field.

Answer

The positive ion experiences a force from left to right. The electron experiences a force from right to left. A positive charge is accelerated in the direction of the electric field, whereas a negative charge is accelerated in the opposite direction.

4B Medium

A proton and an alpha particle are initially at rest in the same uniform electric field. Taking the mass of the alpha particle to be four times the mass of a proton, calculate $\frac{a_{\text{proton}}}{a_{\text{alpha}}}$.

Answer

Since

$$a = \frac{qE}{m}$$

for the proton,

$$a_p = \frac{eE}{m_p}$$

For the alpha particle,

$$a_\alpha = \frac{2eE}{4mp} = \frac{eE}{2mp}$$

Therefore,

$$\frac{a_p}{a_\alpha} = \frac{\frac{eE}{m_p}}{\frac{eE}{2m_p}} = 2$$

$$\boxed{\frac{a_p}{a_\alpha} = 2}$$

4C Hard

An α particle is accelerated through a perpendicular electric field and is deflected with an angle of +2°. If this is now replaced with an Cl^- particle, what is the angle to which it is deflected by? Assume angle $\vartheta \propto \frac{q}{m}$.

Answer

Since

$$\theta \propto \frac{q}{m},$$

we have

$$\frac{\theta_{Cl^-}}{\theta_\alpha} = \frac{\left(\frac{q}{m}\right)_{Cl^-}}{\left(\frac{q}{m}\right)_\alpha}.$$

For the α particle,

$$\left(\frac{q}{m}\right)_\alpha = \frac{+2}{4} = \frac{1}{2}.$$

For the Cl^- ion, taking its relative mass as 35.5,

$$\left(\frac{q}{m}\right)_{Cl^-} = \frac{-1}{35.5}.$$

Hence,

$$\theta_{Cl^-} = 2° \times \frac{\frac{-1}{35.5}}{\frac{1}{2}}$$

$$= 2° \times \left(-\frac{2}{35.5}\right)$$

$$\approx -0.113°.$$

The negative sign indicates that the Cl^- ion is deflected in the opposite direction to the positively charged α particle.

1.5 Distribution of mass and charge within an atom

5A Easy

Describe how mass and charge are distributed within an atom.

Answer

Almost all the mass of an atom is concentrated in its very small nucleus, which contains protons and neutrons. Positive charge is concentrated in the nucleus, while negative charge is distributed through the surrounding electron cloud. Most of the atom's volume is empty space.

5B Medium

In Rutherford's alpha-particle scattering experiment:
1. most alpha particles passed through the foil with little or no deflection,
2. a small fraction were strongly deflected,
3. a very small fraction travelled backwards.

Explain what each observation reveals about atomic structure.

Answer

1. Most particles passed through because atoms are mostly empty space.
2. Strong deflections occurred when alpha particles passed close to a concentrated region of positive charge.
3. Backward scattering showed that almost all the positive charge and mass must be concentrated in a very small, dense nucleus.

5C Hard

Suppose the positive charge of an atom were spread uniformly throughout the entire volume of the atom rather than concentrated in a nucleus. Predict how Rutherford's scattering results would differ.

Answer

An alpha particle passing through a diffuse positive charge would experience many weak forces rather than one very strong force. Most particles would still pass through, and small deflections might occur, but large-angle deflections would be extremely unlikely and backward scattering would essentially not occur. The observed backward scattering therefore supports the existence of a small, concentrated nucleus.

1.6 Relative atomic mass from isotopic abundance

6A Easy

An element consists of 80.0% isotope ^{24}X and 20.0% isotope ^{26}X. Calculate its relative atomic mass.

Answer

$$A_r = \frac{(80.0)(24) + (20.0)(26)}{100} = \frac{1920 + 520}{100}.$$

Therefore,

$$\boxed{A_r = 24.4}$$

6B Medium

Chlorine consists only of 35 Cl and 37 Cl. Its relative atomic mass is 35.45. Calculate the percentage abundance of each isotope.

Answer

Let the fractional abundance of ^{37}Cl be x. The fractional abundance of ^{35}Cl is 1-x.

$$35(1 - x) + 37x = 35.45.$$

Hence,

$$35 + 2x = 35.45,\ x = 0.225.$$

Therefore,

$$\boxed{^{37}\text{Cl} : 22.5\%} \text{ and } \boxed{^{35}\text{Cl} : 77.5\%}$$

6C Hard

An element has three isotopes with mass numbers 24, 25 and 26. The abundance of the mass-25 isotope is twice that of the mass-26 isotope. The relative atomic mass is 24.32. Determine the abundance of each isotope.

Answer

Let the fractional abundance of ^{26}X be x. Then

$$^{25}X : 2x,\ ^{25}\text{X} : 1 - 3x$$

Using the weighted mean,

$$24(1 - 3x) + 25(2x) + 26x = 24.32.$$

Thus,

$$24 - 72x + 50x + 26x = 24.32,$$
$$24 + 4x = 24.32,\ x = 0.08.$$

Therefore,

$$\boxed{^{24}\text{X} : 76\%,\ ^{25}\text{X} : 16\%,\ ^{26}\text{X} : 8\%}.$$

2. Electronic Structure

2.1 Principal shells and subshells

7A Easy

List all the subshells that may exist in the principal shell $n = 3$.

Answer

For $n = 3$, the allowed subshells are
3s, 3p, 3d .

7B Medium

Determine:
1. the number of subshells in the $n = 4$ shell,
2. the total number of orbitals,
3. the maximum number of electrons.

Answer

The $n = 4$ shell contains
4s, 4p, 4d, 4f. There are therefore four subshells. Their numbers of orbitals are

$$1 + 3 + 5 + 7 = 16.$$

Since each orbital holds at most two electrons,

$$16 \times 2 = 32.$$

Thus, the shell contains 4 subshells, 16 orbitals and at most 32 electrons.

7C Hard

Which of the following orbital labels are impossible? 1p, 2d, 3f, 4d, 4f, 5g. For each impossible label, explain why it is not allowed.

Answer

For a shell with principal quantum number n, the allowed values are $\ell = 0, 1, \ldots, n-1$,
where s, p, d, f , g correspond to $\ell = 0, 1, 2, 3, 4$.

- 1p is impossible because $n = 1$ permits only 1s.
- 2d is impossible because $n = 2$ permits only 2s and 2p.
- 3f is impossible because $n = 3$ permits only 3s, 3p and 3d.
- 4d, 4f and 5g are possible.

2.2 Number and relative energies of *s*, *p* and *d* orbitals

8A Easy

State the number of orbitals and the maximum number of electrons in each of the s, p and d subshells.

Answer

Subshell	Number of orbitals	Maximum electrons
s	1	2
p	3	6
d	5	10

8B Medium

Arrange the following orbitals in the approximate order in which they are filled in neutral atoms: 3p, 3d, 4s, 4p, 5s. Explain why orbital energy cannot always be determined using principal quantum number alone.

Answer

The approximate filling order is

$$3\text{p} < 4\text{s} < 3\text{d} < 4\text{p} < 5\text{s} \ .$$

Orbital energy depends not only on principal quantum number but also on penetration and shielding. For example, the 4s orbital penetrates closer to the nucleus than the 3d orbital and is initially lower in energy.

8C Hard — Multi-statement MCQ

Consider the following statements.

1. In a hydrogen atom, orbitals with the same principal quantum number have the same energy.
2. In a many-electron atom, a 3s electron generally experiences greater penetration than a 3p electron.
3. A 4s orbital is always lower in energy than a 3d orbital, regardless of whether the orbitals are occupied.

Which statements are correct?

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

Answer

Statement 1 is correct for a one-electron hydrogen atom. Statement 2 is correct because an s orbital penetrates closer to the nucleus. Statement 3 is incorrect: the relative energies of 4s and 3d change as electrons are added and ions are formed.

$$\boxed{B}$$

2.3 Shapes and orientations of orbitals

9A Easy

Describe the general shapes of an s orbital and a p orbital.

Answer

An s orbital is spherically symmetrical around the nucleus. A p orbital has two lobes on opposite sides of the nucleus and is often described as dumbbell-shaped.

9B Medium

Explain why a p subshell contains three orbitals even though all three orbitals have the same general shape and energy.

Answer

The three p orbitals have different orientations in space: p_x, p_y, p_z. They are oriented along three mutually perpendicular axes. In an isolated atom, the three orbitals normally have the same energy.

9C Hard

Evaluate the following statements:
1. "A p orbital is the path followed by an electron around the nucleus."
2. "The three p orbitals differ mainly in their orientation."
3. "The nucleus lies at a node of a p orbital."

Answer

1. Incorrect. An orbital is a region described by a wavefunction and represents an electron probability distribution. It is not a fixed path.
1. Correct. The three p orbitals have the same general shape but different orientations.
2. Correct. The probability density of a p electron is zero at the nucleus, which lies on the nodal plane.

2.4 Aufbau principle

10A Easy

Use the Aufbau principle to write the ground-state electronic configuration of calcium.

Answer

Calcium has 20 electrons:
$1s^2 2s^2 2p^6 3s^2 3p^4 s^2$ or $[Ar]4s^2$.

10B Medium

A student proposes the configuration $1s^2 2s^2 2p^6 3s^2 3p^5 4s^1$ for the ground state of an 18-electron atom. Explain why this configuration does not represent the ground state and give the correct configuration.

Answer

The $3p$ subshell is lower in energy than the $4s$ subshell and should be filled first. The proposed configuration is an excited-state configuration. The correct ground-state configuration is $1s^2 2s^2 2p^6 3s^2 3p^6$.

10C Hard

The Aufbau principle is sometimes stated as "electrons always enter the lowest-energy orbital available." Explain why this statement is useful but incomplete. Refer to electron–electron repulsion, the closeness in energy of $4s$ and $3d$, and anomalous configurations such as chromium.

Answer

The statement is useful because it predicts the general order in which orbitals are occupied. It is incomplete because orbital energies depend on electron–electron interactions, the energies of $4s$ and $3d$ are very close, and orbital energies change as orbitals become occupied. Configurations such as chromium reflect the total energy of the whole atom rather than a rigid filling sequence:

$$\text{Cr} : [\text{Ar}]\ 3d^5 4s^1 \text{ rather than the simplest prediction } [\text{Ar}]\ 3d^4 4s^2\ .$$

2.5 Pauli exclusion principle

11A Easy

State the Pauli exclusion principle and explain why one orbital can contain no more than two electrons.

Answer

No two electrons in the same atom can have the same set of four quantum numbers. Consequently, one orbital can contain at most two electrons, and those two electrons must have opposite spins.

11B Medium

Identify the error in each situation:
1. one orbital contains three electrons,
2. one orbital contains two electrons with parallel spins,
3. two electrons in the same atom are assigned an identical set of four quantum numbers.

Answer

All three situations violate the Pauli exclusion principle.
1. An orbital can hold at most two electrons.
2. Two electrons in the same orbital must have opposite spins.
3. No two electrons in one atom may have identical sets of four quantum numbers.

11C Hard

Classify each configuration as violating the Aufbau principle, the Pauli exclusion principle, Hund's rule, or none of these.
1. Two electrons of identical spin occupy the same orbital.
2. Two $2p$ electrons are paired in one orbital while the other $2p$ orbitals are empty.
3. An electron occupies $4s$ while the $3p$ subshell is incomplete.
4. Five $3d$ electrons occupy five separate orbitals with parallel spins.

Answer

1. Pauli exclusion principle.
2. Hund's rule.
3. Aufbau principle.
4. None; this is the ground-state arrangement of a d^5 subshell.

2.6 Hund's rule

12A Easy

Draw the orbital-box representation of a $2p^2$ configuration in its ground state.

Answer

The two electrons occupy separate orbitals with parallel spins: ↑ ↑ Pairing them in one orbital would not give the ground-state arrangement.

12B Medium

Determine the number of unpaired electrons in carbon, nitrogen, oxygen and fluorine. Explain how Hund's rule is used.

Answer

Considering their $2p$ configurations:

- Carbon, $2p^2$: 2 unpaired electrons.
- Nitrogen, $2p^3$: 3 unpaired electrons.
- Oxygen, $2p^4$: 2 unpaired electrons.
- Fluorine, $2p^5$: 1 unpaired electron.

Hund's rule states that electrons occupy degenerate orbitals singly with parallel spins before pairing occurs.

12C Hard

Two possible arrangements for a p^3 configuration are proposed:

- Arrangement A: one electron in each p orbital with parallel spins.
- Arrangement B: two electrons paired in one orbital and one electron in a second orbital.

Explain why Arrangement A is lower in energy. Your explanation should go beyond simply quoting Hund's rule.

Answer

Arrangement A minimizes electron pairing within individual orbitals. In Arrangement B, two electrons occupy the same region of space, increasing electron–electron repulsion. Arrangement A also receives stabilization associated with parallel-spin electrons occupying separate degenerate orbitals. Therefore, A has the lower total energy.

2.7 Electronic configurations of atoms and ions

13A Easy

Write the electronic configurations of:

1. S
2. S^{2-}
3. Na^{+}

Answer

$$\begin{aligned} S: &\quad 1s^2 2s^2 2p^6 3s^2 3p^4 \\ S^{2-}: &\quad 1s^2 2s^2 2p^6 3s^2 3p^6 \\ Na^{+}: &\quad 1s^2 2s^2 2p^6 \end{aligned}$$

13B Medium

Write the electronic configurations of:

1. Fe,
2. Fe^{2+}
3. Fe^{3+}
4. Cu^{2+}

Answer

$$\begin{aligned} Fe: &\quad [\mathrm{Ar}]3d^6 4s^2 \\ Fe^{2+}: &\quad [\mathrm{Ar}]3d^6 \end{aligned}$$

Fe^{3+} : [Ar]$3d^5$
Cu^{+} : [Ar]$3d^{10}$

The $4s$ electrons are removed before the $3d$ electrons.

13C Hard

A species contains 23 protons and 18 electrons.
1. Identify the species.
2. Write its full and abbreviated electronic configurations.
3. State the number of unpaired electrons.
4. Explain whether it is isoelectronic with argon.

Answer

The element with 23 protons is vanadium. Its charge is $23 - 18 = 5+$, so the species is V^{5+} . Its configuration is $1s^2 2^2 2p^6 3s^2 3p^6$ or $[Ar]$. It has no unpaired electrons and is isoelectronic with argon because both species contain 18 electrons.

2.8 Removal of $4s$ electrons before $3d$ electrons

14A Easy

Write the electronic configurations of Fe and Fe^{2+} . Indicate which electrons are removed during ion formation.

Answer

The two $4s$ electrons are removed first:
Fe : [Ar]$3d^6 4s^2$,
Fe^{2+} : [Ar]$3d^6$,

14B Medium

Write the electronic configurations of:
1. Mn
2. Mn^{2+}
3. Mn^{3+}

State the order in which the electrons are removed.

Answer

Mn : [Ar]$3d^5 4s^2$.
Removing the two $4s$ electrons gives
Mn^{2+} : [Ar]$3d^5$.
Removing one additional $3d$ electron gives
Mn^{3+} : [Ar]$3d^4$.

14C Hard

A student argues:
“Because $4s$ fills before $3d$, $3d$ electrons must always be removed before $4s$ electrons.” Explain the flaw in this argument. Discuss how orbital energies change once the $3d$ subshell becomes occupied.

Answer

The order in which orbitals fill does not necessarily determine the order in which electrons are removed. Before the $3d$ subshell is occupied, $4s$ is slightly lower in energy and fills first. Once $3d$ electrons are present, electron–

electron interactions alter the orbital energies. The $4s$ orbital becomes higher in energy and more spatially extended than the $3d$ orbitals. Therefore, $4s$ electrons are removed first when transition-metal ions form.

2.9 Stability of half-filled and fully filled subshells

15A Easy

Write the observed ground-state configurations of chromium and copper. State how each differs from the simplest Aufbau prediction.

Answer

Chromium:
$\boxed{[\mathrm{Ar}]3d^5 4s^1}$
rather than
$[\mathrm{Ar}]3d^4 4s^2$.
Copper:
$\boxed{[\mathrm{Ar}]3d^{10} 4s^1}$
rather than
$[\mathrm{Ar}]3d^9 4s^2$.

15B Medium

Compare the two possible configurations for chromium:
$[\mathrm{Ar}]3d^4 4s^2$ and $[\mathrm{Ar}]3d^5 4s^1$. Explain why the second configuration is favoured.

Answer

The $3d$ and $4s$ orbitals are very close in energy. Promoting one $4s$ electron produces a half-filled $3d^5$ subshell in which every d orbital is singly occupied, pairing repulsion is reduced, and parallel-spin stabilization is maximized. These stabilizing effects slightly exceed the energy required to promote the electron.

15C Hard

An unfamiliar atom is predicted by the simplest Aufbau treatment to have the configuration

$$[\text{core}]\ (n-1)d^9 ns^2$$

Explain what alternative configuration might occur. Discuss the balance between promotion energy, reduced pairing repulsion and stabilization of a completely filled d subshell.

Answer

A possible alternative is

$$[\text{core}]\ (n-1)d^{10} ns^1$$

One ns electron is promoted into the d subshell. This may be favoured because the d and s subshells are close in energy, a completely filled d^{10} subshell is especially stable, and the change can reduce the total electron-pairing and interaction energy. The exact configuration depends on the total energy of the atom rather than on a rigid filling rule.

3. Ionisation Energy

3.1 Definition of first and successive ionisation energies

16A Easy

Define the first ionisation energy of an element. Your definition must include the appropriate physical state, amount of substance and type of species formed.

Answer

The first ionisation energy is the energy required to remove one electron from each atom in one mole of gaseous atoms to form one mole of gaseous singly charged positive ions: $X(g) \rightarrow X^+(g) + e^-$

16B Medium

Write equations, including state symbols, for the first, second and third ionisation of magnesium.

Answer

First ionisation:
$Mg(g) \rightarrow Mg^+(g) + e^-$
Second ionisation:
$Mg^+(g) \rightarrow Mg^{2+}(g) + e^-$
Third ionisation:
$Mg^{2+}(g) \rightarrow Mg^{3+}(g) + e^-$.

16C Hard

The first and second ionisation energies of magnesium are

$$\text{IE}_1 = 738 \text{ kJ/mol and IE}_2 = 1451 \text{ kJ/mol.}$$

Calculate the energy required to convert 0.250 mol of gaseous magnesium atoms completely into gaseous Mg^{2+} ions.

Answer

The energy required per mole is
$738 + 1451 = 2189$ kJ/mol.
For 0.250 mol,
$E = 0.250(2189) = 547.25$ kJ.
To three significant figures,
$E = 547$ kJ .

3.2 Nuclear charge

17A Easy

Explain why the first ionisation energy of magnesium is greater than that of sodium.

Answer

Both elements lose an electron from the 3*s* subshell, so shielding and distance from the nucleus are similar. Magnesium has one more proton than sodium, so its greater nuclear charge attracts the outer electron more strongly. More energy is therefore required to remove it.

17B Medium

Arrange the following isoelectronic species in increasing ionisation energy:

$$O^{2-}, F^-, Na^+, Mg^{2+}.$$

Explain your reasoning in terms of nuclear charge.

Answer

All four species contain 10 electrons, but their proton numbers are 8, 9, 11 and 12, respectively. A greater nuclear charge attracts the same number of electrons more strongly. Therefore,

$$\boxed{O^{2-} < F^{-} < Na^{+} < Mg^{2+}.}$$

17C Hard

Compare the energy required for the processes

$$B^{+}(g) \longrightarrow B^{2+}(g) + e^{-}$$

and

$$C^{+}(g) \longrightarrow C^{2+}(g) + e^{-}.$$

Account for both nuclear charge and the electronic configurations of the ions involved.

Answer

The relevant configurations are

$$B^{+}: \quad 1s^2 2s^2$$

and

$$C^{+}: \quad 1s^2 2s^2 2p^1.$$

Removing an electron from B^{+} requires removing a $2s$ electron, whereas removing an electron from C^{+} removes a higher-energy, less penetrating and more strongly shielded $2p$ electron. Although carbon has the greater nuclear charge, the subshell effect is sufficiently important that removing the $2p$ electron from C^{+} is slightly easier than removing a $2s$ electron from the filled $2s^2$ subshell of B^{+}.

3.3 Shielding

18A Easy

Explain why first ionisation energy generally decreases from lithium to sodium to potassium.

Answer

Moving down the group, the outer electron occupies a shell farther from the nucleus and is shielded by more inner shells. These effects weaken the attraction between the outer electron and the nucleus and outweigh the increase in nuclear charge.

18B Medium

A $2s$ electron and a $2p$ electron are in the same principal shell. Explain why the $2s$ electron generally experiences greater penetration, less effective shielding and stronger attraction to the nucleus.

Answer

A $2s$ orbital penetrates more deeply into the region occupied by the $1s$ electrons. The $2s$ electron therefore spends more time close to the nucleus and inside part of the inner-shell electron density. It experiences a greater effective nuclear charge and is less effectively shielded than a $2p$ electron.

18C Hard — **Multi-statement MCQ**

Consider the following statements.

1. Inner-shell electrons generally shield outer-shell electrons more effectively than electrons in the same shell.

1. An s electron generally penetrates closer to the nucleus than a p electron with the same principal quantum number.

1. Each inner electron reduces the nuclear attraction experienced by an outer electron by exactly one proton charge. Which statements are correct?

A. 1 only B. 1 and 2 only C. 2 and 3 only D. 1 and 3 only E. 1, 2 and 3

Answer

Statements 1 and 2 are correct. Statement 3 is incorrect because shielding is neither exact nor complete; each inner electron does not cancel precisely one unit of nuclear charge.

B

3.4 Distance from the nucleus

19A Easy

Explain why the first ionisation energy of potassium is lower than that of sodium.

Answer

Potassium loses a 4*s* electron, whereas sodium loses a 3*s* electron. The 4*s* electron is, on average, farther from the nucleus and is more strongly shielded by inner electrons. It is therefore less strongly attracted to the nucleus and easier to remove.

19B Medium

Suppose a 3*s* electron and a 4*s* electron experience approximately the same effective nuclear charge. Which electron would be easier to remove? Explain using average distance from the nucleus.

Answer

The 4*s* electron is easier to remove because it is, on average, farther from the nucleus. Electrostatic attraction becomes weaker as electron–nucleus distance increases.

19C Hard

A hydrogen atom is promoted from its ground state to an excited state in which its electron occupies a shell with a larger principal quantum number.

Explain how this affects:

1. the average electron–nucleus distance,
2. the attraction between the electron and nucleus,
3. the ionisation energy.

Answer

In the excited state, the average electron–nucleus distance increases and the electrostatic attraction becomes weaker. The electron also has a higher energy and lies closer to the ionisation limit. Consequently, less additional energy is required to remove it, so the ionisation energy decreases.

3.5 Electron–electron repulsion

20A Easy

Explain why removing an electron from a negatively charged ion is generally easier than removing one from the corresponding neutral atom.

Answer

A negative ion contains more electrons but the same number of protons. The additional electron increases electron–electron repulsion and shielding, reducing the effective attraction experienced by each outer electron. An electron is therefore generally easier to remove from the negative ion.

20B Medium

Compare the attraction experienced by the electron in He^+ with that experienced by either electron in neutral helium. Explain the role of electron–electron repulsion.

Answer

In He^+ , the single electron experiences the attraction of the +2 nucleus without repulsion or shielding from another electron. In neutral helium, the two electrons repel and partially shield one another. The electron in He^+ is therefore held more strongly.

20C Hard

The second ionisation energy of an element is always greater than its first ionisation energy. Explain why, referring to ionic charge, electron–electron repulsion and attraction between the remaining electrons and the nucleus.

Answer

After the first electron is removed, the species becomes positively charged while the number of protons remains unchanged. There is one fewer electron, so electron–electron repulsion and shielding decrease. The remaining electrons experience stronger attraction to the nucleus, and more energy is required to remove the second electron.

3.6 Subshell effects

21A Easy

Explain why the first ionisation energy of aluminium is lower than that of magnesium despite aluminium having a greater nuclear charge.

Answer

Magnesium has the configuration $[Ne]3s^2$, whereas aluminium has $[Ne]3s^23p^1$. The electron removed from aluminium is a $3p$ electron. It is higher in energy, less penetrating and more strongly shielded than a $3s$ electron. It is therefore easier to remove despite aluminium's greater nuclear charge.

21B Medium

Explain why the first ionisation energy of boron is lower than that of beryllium. Compare the energies and penetration of the $2s$ and $2p$ subshells.

Answer

Beryllium has the configuration $1s^22s^2$, whereas boron has $1s^22s^22p^1$. The first electron removed from boron is a $2p$ electron, which is higher in energy and less penetrating than a $2s$ electron. Hence, boron has a lower first ionisation energy despite its greater nuclear charge.

21C Hard

The first ionisation energy of gallium is not dramatically lower than that of aluminium, even though gallium lies below aluminium in Group 13. Suggest an explanation involving the relatively poor shielding provided by the filled $3d$ subshell.

Answer

Gallium has a filled $3d^{10}$ subshell between its valence shell and nucleus. The $3d$ electrons shield the nuclear charge relatively poorly. Consequently, the $4p$ electron in gallium experiences a larger effective nuclear charge than might otherwise be expected. This offsets much of the effect of the additional shell and increased distance from the nucleus.

3.7 Paired-electron repulsion

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22A Easy

Explain why the first ionisation energy of oxygen is lower than that of nitrogen.

Answer

Nitrogen has a $2p^3$ configuration with one electron in each p orbital. Oxygen has a $2p^4$ configuration, so one p orbital contains a pair of electrons. Repulsion between the paired electrons makes one of them easier to remove, outweighing the increase in nuclear charge from nitrogen to oxygen.

22B Medium

Predict and explain the corresponding anomaly between phosphorus and sulfur.

Answer

Phosphorus has $3p^3$, whereas sulfur has $3p^4$. In sulfur, one $3p$ orbital contains a pair of electrons. Paired-electron repulsion makes one electron easier to remove.
Therefore,

$$\boxed{\mathrm{IE}_1(\mathrm{S}) < \mathrm{IE}_1(\mathrm{P})},$$

despite sulfur having the greater nuclear charge.

22C Hard

Consider four hypothetical atoms whose outer configurations are

$$np^1, np^2, np^3, np^4.$$

Assume that the electrons experience approximately the same effective nuclear charge and belong to the same principal shell. Predict which configuration would show an unexpectedly low ionisation energy relative to the preceding configuration. Explain using orbital occupancy and pairing repulsion.

Answer

The unexpectedly low value occurs for np^4 . The np^3 configuration has one electron in each of the three p orbitals. The fourth electron in np^4 must pair with another electron. Repulsion within that occupied orbital makes the paired electron easier to remove.

3.8 Identifying groups and periods from successive ionisation energies

23A Easy

An element in Period 3 has the following successive ionisation energies: 738, 1451, 7733, 10540 kJ/mol.
1. Identify the major discontinuity.
2. Determine the group of the element.
3. Identify the element.

Answer

There is a large jump between the second and third ionisation energies. This indicates that the atom has two valence electrons; after two electrons are removed, the next electron comes from an inner shell. The element is therefore in Group 2. The Period 3 Group 2 element is Mg .

23B Medium

A successive-ionisation-energy plot shows major jumps after removal of the first and ninth electrons.
Determine:
1. the total number of electrons in the atom,
2. its electronic configuration,

3. its group and period,
4. its likely identity.

Answer

The first major jump after one electron indicates one valence electron. The second major jump after nine electrons means that, after removing one outer-shell electron and eight electrons from the next shell, only the $1s2$ electrons remain. The atom therefore contains

$$1 + 8 + 2 = 11$$

electrons. Its configuration is $1s^2, 2s^2, 2p^6, 3s^1$. It is in Group 1 and Period 3, so the element is Na .

23C Hard

An unknown main-group atom shows:

- a major jump after the seventh electron is removed,
- another major jump after the fifteenth electron is removed.

Deduce the number of valence electrons, the total number of electrons, the full electronic configuration, the group and period, and the identity of the element.

Answer

The first jump after seven electrons indicates seven valence electrons. The next eight electrons belong to the second shell:

$$15 - 7 = 8.$$

After 15 electrons have been removed, the $1s2$ electrons remain. Thus the total number of electrons is

$$15 + 2 = 17.$$

The configuration is $1s^2 2s^2 2p^6 3s^2 3p^5$. The element is in Group 17 and Period 3 and is therefore Cl .

3.9 Explaining apparent anomalies in periodic trends

24A Easy

Across Period 3, first ionisation energy generally increases. Explain why the first ionisation energy of aluminium is nevertheless lower than that of magnesium.

Answer

The electron removed from aluminium is a $3p$ electron, whereas the electron removed from magnesium is a $3s$ electron. The $3p$ electron is higher in energy, less penetrating and more strongly shielded, so it is easier to remove.

24B Medium

Explain why nitrogen has a higher first ionisation energy than oxygen even though oxygen has the greater nuclear charge.

Answer

Nitrogen has the half-filled configuration $2p^3$. Oxygen has $2p^4$, so one $2p$ orbital contains a pair of electrons. Repulsion between the paired electrons makes one electron easier to remove from oxygen.

24C Hard

Across Period 2, first ionisation energy generally increases. Identify and explain the two principal exceptions. For each exception, state whether subshell energy or paired-electron repulsion is the dominant explanation, and explain why the general increasing trend still remains valid.

Answer

The two exceptions are

$$\text{IE1(B)} < \text{IE1(Be)}$$

and

$$\text{IE1(O)} < \text{IE1(N)}.$$

For beryllium and boron,

$$\text{Be} : 1s^2 2s^2, \text{ B: } 1s^2 2s^2 2p^1$$

The $2p$ electron in boron is higher in energy and more strongly shielded than a $2s$ electron. This is mainly a subshell-energy effect.
For nitrogen and oxygen,

$$\text{N} : 2p^3, \text{ O: } 2p^4$$

Oxygen contains a pair of electrons in one $2p$ orbital. Paired-electron repulsion makes one electron easier to remove. This is mainly a pairing-repulsion effect. The general trend remains valid because nuclear charge increases across the period while shielding changes relatively little.

4. Olympiad Extension: Mass Spectrometry

The following questions provide the necessary principles in their stems. Students are assessed on interpretation, deduction and calculation rather than prior knowledge of instrumental design.

4.1 Reading an isotope spectrum

MS1 Reading an isotope spectrum

A mass spectrum separates singly charged ions according to their mass-to-charge ratio, $\frac{m}{z}$. For singly charged ions, $\frac{m}{z}$ is numerically equal to the isotope's relative mass.
The spectrum of element X contains the following peaks:

$\frac{m}{z}$	Relative intensity
24	79
25	10
26	11

1. State the number of naturally occurring isotopes shown.
2. Calculate the relative atomic mass of X.
3. Suggest the identity of X.

Answer

Three isotopes are shown. Since the relative intensities total 100, they may be treated as percentages:

$$A_r = \frac{24(79) + 25(10) + 26(11)}{100} = \frac{1896 + 250 + 286}{100} = 24.32.$$

Thus,

$$\boxed{A_r = 24.32}$$

An element with relative atomic mass about 24.3 and isotopes 24, 25 and 26 is magnesium:

$$\boxed{X = \text{Mg}}$$

4.2 Time-of-flight reasoning

MS2 Time-of-flight reasoning

In a time-of-flight mass spectrometer, ions of charge q are accelerated from rest through the same potential difference V. The electrical energy gained becomes kinetic energy:

$$qV = \tfrac{1}{2}mv^2.$$

The time required to travel through a field-free tube of length L is

$$t = \frac{L}{v}.$$

1. Derive an expression for t in terms of m, q, V and L.
2. Two singly charged isotopes have masses 25 and 100. Determine the ratio of their flight times.
3. Explain why heavier ions arrive later.

Answer

From

$$qV = \tfrac{1}{2}mv^2,$$

we obtain

$$v = \sqrt{\tfrac{2qV}{m}}.$$

Therefore,

$$\boxed{t = L\sqrt{\tfrac{m}{2qV}}}.$$

For ions with the same charge,

$$t \propto \sqrt{m}.$$

Hence,

$$\tfrac{t_{100}}{t_{25}} = \sqrt{\tfrac{100}{25}} = 2,$$

so

$$\boxed{t_{100} : t_{25} = 2 : 1}.$$

Heavier ions receive the same electrical energy but attain lower speeds because the same kinetic energy is associated with a larger mass.

4.3 The meaning of mass-to-charge ratio

MS3 The meaning of m/z

A mass spectrometer does not measure mass directly. It measures the mass-to-charge ratio m/z , where z is the number of positive elementary charges on the ion. Two peaks are produced by $^{20}Ne^{+}$ and $^{40}Ar^{2+}$.

1. Calculate the m/z value of each ion.
2. Explain why the two ions appear at the same position.
3. Explain why a single peak at $m/z = 20$ is not sufficient evidence that an ion has a mass of 20.

Answer

For $^{20}Ne^{+}$,

$$\frac{m}{z} = \frac{20}{1} = 20.$$

For $^{40}Ar^{2+}$,

$$\frac{m}{z} = \frac{40}{2} = 20.$$

Both therefore appear at

$$\boxed{\frac{m}{z} = 20}.$$

The instrument measures a ratio rather than mass alone. A peak at $\frac{m}{z} = 20$ could represent a mass-20 singly charged ion, a mass-40 doubly charged ion, or any other ion with the same ratio.

4.4 Chlorine isotope patterns

MS4 Chlorine isotope patterns

Chlorine has two common isotopes:

- ^{35}Cl, abundance 75%
- ^{37}Cl, abundance 25%

A Cl_2 molecule can therefore contain two ^{35}Cl atoms, one of each isotope, or two ^{37}Cl atoms.

Determine:

1. the masses of the three possible Cl_2^{+} molecular ions
2. their expected relative probabilities,
3. the approximate ratio of their peak intensities.

Answer

The possible masses are

$$35+35=70, \quad 35+37=72, \quad 37+37=74.$$

The probabilities are

$$P(70)=0.75^2=0.5625,$$
$$P(72)=2(0.75)(0.25)=0.375,$$
$$P(74)=0.25^2=0.0625.$$

Therefore,

$$0.5625:0.375:0.0625=\boxed{9:6:1}.$$

The peaks at $\frac{m}{z}=70,72,74$ appear in the approximate ratio $9:6:1$.

4.5 Distinguishing isotope abundance from instrumental response

MS5 Evaluating a conclusion

A mass spectrum shows two peaks at $m/z=79$ and $m/z=81$ with nearly equal intensities.

A student concludes:

"The two isotopes must have exactly equal natural abundances." Evaluate this conclusion.

Answer

The conclusion is plausible but stronger than the evidence warrants. Peak intensity can depend on natural abundance, ionisation efficiency, ion transmission, detector response and possible overlap with other ions. Because isotopes of the same element usually have nearly identical chemical behaviour, similar intensities are strong evidence for similar abundances. However, exact abundances should be obtained after appropriate calibration and correction for instrumental effects.

4.6 Instrumental resolution

MS6 Instrumental resolution

The resolving power R of a mass spectrometer is defined as

$$R=\frac{m}{\Delta m},$$

where m is the approximate mass and Δm is the smallest mass difference that can be distinguished. Two ions have masses 28.00 and 28.03.

1. Calculate the minimum resolving power needed to separate them.
2. State whether an instrument with $R=500$ can distinguish the peaks.
3. State whether an instrument with $R=1200$ can distinguish the peaks.

Answer

$$\Delta m=28.03-28.00=0.03.$$

Using $m\approx 28.00$,

$$R=\frac{28.00}{0.03}=933.3.$$

The minimum whole-number resolving power is therefore approximately

$$\boxed{R=934}.$$

An instrument with $R=500$ cannot resolve the peaks, whereas an instrument with $R=1200$ can.

5. Olympiad Extension: Photoelectron Spectroscopy

In photoelectron spectroscopy (PES), photons eject electrons from gaseous atoms. The energy relationship is

$$E_{\text{photon}}=E_{\text{binding}}+E_{\text{kinetic}}.$$

A simplified PES spectrum is interpreted as follows:

- peak position indicates electron binding energy;
- peak area is proportional to the number of electrons in the corresponding subshell;
- a higher binding energy means that an electron is held more strongly.

The questions below include all information needed for reasoning.

5.1 Identifying subshells

PES1 Identifying subshells

An atom produces three PES peaks:

Peak	Relative binding energy	Relative area
A	very high	2
B	medium	2
C	low	1

1. Assign each peak to a subshell.
2. Write the electronic configuration.
3. Identify the element.

Answer

The highest-binding-energy electrons are closest to the nucleus. Therefore,

$$A : 1s^2, B : 2s^2, C : 2p^1$$

The configuration is

$$1s^2, 2s^2, 2p^1.$$

The atom contains five electrons and is therefore

$$\boxed{\text{B}}$$

5.2 Using peak areas

PES2 Using peak areas

An unknown atom has five PES peaks with relative areas

$$2 : 2 : 6 : 2 : 3.$$

The peaks are listed from highest to lowest binding energy.

1. Assign each peak to a subshell.
2. Determine the total number of electrons.
3. Identify the atom.
4. State the number of unpaired electrons in its ground state.

Answer

The peaks correspond to

$$1s^2, \quad 2s^2, \quad 2p^6, \quad 3s^2, \quad 3p^3.$$

The total number of electrons is

$$2 + 2 + 6 + 2 + 3 = 15.$$

The atom is phosphorus:

$$\boxed{\text{P}}.$$

Its $3p^3$ electrons occupy separate orbitals according to Hund's rule, so it has

$$\boxed{\text{3 unpaired electrons}}.$$

5.3 The magnesium–aluminium anomaly

PES3 Explaining the magnesium–aluminium anomaly

Simplified PES data for the valence electrons of magnesium and aluminium are shown below.

Element	Valence subshell	Relative binding energy
Mg	$3s$	higher
Al	$3p$	lower

Both elements contain inner electrons with similar shielding.
1. Which element should have the lower first ionisation energy?
2. Explain why.
3. Explain why this does not contradict aluminium having the greater nuclear charge.

Answer

Aluminium should have the lower first ionisation energy. The electron removed from aluminium is a $3p$ electron. Compared with a $3s$ electron, it is less penetrating, more strongly shielded, higher in energy and more weakly bound. Although aluminium has the greater nuclear charge, the subshell effect outweighs the increase in nuclear attraction. Therefore,

$$\boxed{IE_1(Al) < IE_1(Mg)\ .}$$

5.4 Comparing the same subshell across a period

PES4 Binding-energy shifts across a period

Two elements X and Y have valence configurations

$$X : 3s^1,\ Y : 3s^2.$$

The $3s$ PES peak of Y occurs at a higher binding energy than that of X.
1. Which element has the greater nuclear charge?
2. Identify X and Y if they are consecutive Period 3 elements.
3. Explain the shift in binding energy.

Answer

Element Y has the greater nuclear charge. The consecutive Period 3 elements are

$$X = \text{Na},\ Y = \text{Mg}.$$

The two atoms have similar shielding because the additional electron enters the same principal shell. Magnesium has one additional proton, increasing the effective nuclear attraction on the $3s$ electrons. Its $3s$ peak therefore occurs at higher binding energy.

5.5 Paired-electron repulsion

PES5 Paired-electron repulsion

Nitrogen and oxygen have the valence configurations

$$\text{N} : 2p^3,\ \text{O} : 2p^4$$

. In oxygen, one $2p$ orbital contains two electrons.
1. Why is one of oxygen's $2p$ electrons easier to remove than might be expected from the increase in nuclear charge?
1. How does this explain the first-ionisation-energy anomaly between nitrogen and oxygen?
2. Would the total area of oxygen's $2p$ peak be greater or smaller than nitrogen's?

Answer

The paired electrons in one oxygen $2p$ orbital repel one another. This raises the energy of the paired electrons and makes one easier to remove. The pairing repulsion outweighs the increase in nuclear charge, so

$$\boxed{IE_1(O) < IE_1(N)\ .}$$

Oxygen has four $2p$ electrons whereas nitrogen has three, so oxygen's total $2p$ peak area is greater.

5.6 Identifying an ion from its spectrum

PES6 Identifying an ion

A gaseous species produces PES peaks corresponding to $1s^2 2s^2 2^6$. The nucleus of the species contains 12 protons.
1. Determine the number of electrons.
2. Determine the charge of the species.
3. Identify the species.
4. Predict whether its electrons would have higher or lower binding energies than those of neutral neon.

Answer

The spectrum shows

$$2 + 2 + 6 = 10$$

electrons. Since the nucleus contains 12 protons, the charge is

$$12 - 10 = 2 + .$$

The species is Mg^{2+} . Both Mg^{2+} and neon contain 10 electrons, but Mg^{2+} has two more protons. Its electrons therefore experience stronger nuclear attraction and have higher binding energies.

5.7 Distinguishing ground and excited states

PES7 Ground and excited states

An atom normally has the ground-state configuration $1s^2 2s^2 2p^6 3s^2$. After absorbing energy, one electron is promoted, producing $1s^2 2s^2 2p^6 3s^1 3p^1$. A simplified PES experiment is carried out before the atom relaxes.
1. How would the number of occupied valence subshells change?
2. Which electron would have the lower binding energy: the $3s$ or $3p$ electron?
3. Would the excited atom be easier or harder to ionise than the ground-state atom?

Answer

The ground state has only the $3s$ valence subshell occupied, whereas the excited state has both $3s$ and $3p$ occupied. The $3p$ electron has the lower binding energy because it is higher in energy, less penetrating and more strongly shielded. The excited atom is easier to ionise because the promoted electron already has higher energy and requires less additional energy for removal.

5.8 Comparing mass spectrometry and PES

PES8 Comparing what MS and PES reveal

Two experimental results are obtained for an unknown element.
Mass spectrum: the element has isotopes with masses 34, 36 and 38.
PES spectrum: the peak areas correspond to
$1s^2 2s^2 2p^6 3s^2 3p^4$.
1. What information does the mass spectrum provide?
2. What information does the PES spectrum provide?
3. Identify the element.
4. Explain why the mass spectrum alone cannot determine the ground-state electronic configuration.

Answer

The mass spectrum provides information about isotopic masses and relative abundances. The PES spectrum provides information about occupied subshells, electron numbers and binding energies. The PES configuration contains

$$2 + 2 + 6 + 2 + 4 = 16$$

electrons. The element with proton number 16 is sulfur:

$$\boxed{S}$$

The masses 34, 36 and 38 are therefore isotopes of sulfur. The mass spectrum alone does not show how electrons are distributed among shells and subshells. It therefore cannot determine the ground-state electronic configuration by itself.